MedGuard: Full Agentic Pharma Cold-Chain Resilience Platform

Complete architecture — all agents restored, all functioning, built to win

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0. What changed from the trimmed version, and why it's back

Earlier I cut your ~20-agent architecture down to 5 to reduce risk. You're right to push back: the 5 are the *compulsory floor*, not a ceiling, and if your goal is to win and to build something genuinely useful, the extra agents are exactly what makes this stand out from every other team's "we built the 5 required agents" submission.

So here's the real fix — not fewer agents, but **honest depth per agent**. Every agent below is either:

- **Full-depth** — real logic, real APIs, this is what you polish and demo live, or
- **Working-but-simple** — real logic, real data structures, genuinely functions, just with simpler models/rules than a production system, or
- **Minimum-viable-real** — a real, callable version that does the actual job on a narrow slice (one country, one scenario, one query), clearly scoped rather than faked.

Nothing here is a label with no code behind it. That's the difference between "we have 20 boxes on a slide" (what most teams do, and what judges are trained to smell) and "we have 20 agents and can show you any of them working" (what wins).

One-line pitch, updated:

> "MedGuard is a full-stack agentic resilience platform for pharma cold-chain logistics — it senses disruptions (verified, not rumor-based), predicts cascading impact, simulates what-if scenarios on a digital twin, reroutes shipments using live traffic/weather data, automatically handles cross-border regulatory paperwork, rebalances inventory, and keeps working offline at the last mile where connectivity — and medicine access — are weakest. Every high-stakes decision still needs a human's approval."

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1. The problem, stated precisely

- Vaccines and biologics require **2–8°C** continuously. A few hours outside range is a **destruction event**, not a delay.
- The last mile to rural clinics is exactly where **connectivity is worst and monitoring is currently weakest** — the opposite of where cloud-only "AI supply chain" tools focus.
- A single failure cascades: a warehouse power cut becomes a cold-storage failure, becomes a temperature excursion, becomes a batch quarantine, becomes a distributor shortage, becomes a hospital stockout, becomes a patient-level access problem. Most tools only look at the first link.
- Global shocks (Red Sea rerouting, Strait of Hormuz disruption, tariff volatility, port congestion) are now the norm, not the exception — disruption is a continuous condition, not a rare edge case.
- Rerouting a pharma shipment isn't just a geography problem — the instant it crosses into a new jurisdiction, different import/permit rules apply, and existing tools treat that as somebody's manual follow-up work.

The scenario you pitch: a pharma cold-chain shipment moving from a regional hub through a disruption-prone corridor to an under-served, low-connectivity clinic — and everything that can go wrong, does, live, in your demo.

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2. Design principles (say these out loud to judges)

1. **Data stays with its owner.** Suppliers/logistics/warehouses expose narrow, authorized answers via their own agent — never raw database access.
2. **Offline is the default state, not an error state.** The last-mile edge layer keeps sensing, deciding, and logging with zero connectivity, and reconciles automatically once it's back.
3. **Deterministic tools decide what's *allowed*; models decide what's *likely*.** Compliance/sanctions/safety thresholds are rule-based and auditable — never "the LLM decided this was legal." This directly addresses a documented weakness of agentic LLM systems (see Section 8) — say so explicitly, it signals maturity.
4. **Risk-tiered autonomy.** Low-risk actions auto-execute, medium needs a one-click confirm, high-risk (new jurisdiction, new supplier, big spend, pending mandatory permit) requires a named human approver.
5. **Every agent must be demoable, not just describable.** If you can't click a button and watch it run, it's a slide, not a system — cut scope on model sophistication before you cut scope on "does this actually execute."

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3. Full agent roster — all layers

TIER 0 — The five compulsory agents (per the official brief; build these to full depth, they are what's graded)

Agent	Official job	What we build	Useful free API
Disruption Sensing Agent	Monitor maritime/weather/tariff/ geopolitical/port signals, classify severity, escalate ambiguous signals to a human	Ingests weather, disaster feeds, conflict/protest data, broad news — filtered through the **Disruption Verification Layer** (Section 6) so nothing unverified reaches a decision. Also ingests the shipment's own temperature/GPS stream.	**Open-Meteo** (weather forecast, no key: https://open-meteo.com/) + **GDACS** (disaster alerts: https://www.gdacs.org/xml/rss.xml) + **ACLED** (conflict/protest events: https://acleddata.com/) + **GDELT** (broad news events: https://www.gdeltproject.org/data.html)
Scenario Planning Agent	Run "what-if" scenarios (reroute, tariff spike, supplier offline) and evaluate against cost/time/risk	Generates 3–5 candidate responses (wait / reroute / transfer inventory / expedite / alternate supplier), scores each using outputs from Impact Prediction + Demand Forecasting (Tier 2), and hands ranked	None needed — this is internal computation over data already gathered by other agents.

		options to Optimization.	
Rerouting & Logistics Agent	Execute the chosen response: adjust routes, rebalance freight, update timelines, notify downstream partners	Houses the **Rerouting Intelligence** module (Section 5): live traffic + weather + verified unrest scoring picks the actual road/route, then this agent executes the change and pushes it to the Dashboard Timeline.	**TomTom Routing + Traffic API** (real live traffic, free tier: https://developer.tomtom.com/traffic-api/documentation) or **OpenRoute Service** (free route/ETA, no live traffic: https://openrouteservice.org/dev/#/api-docs)
Inventory Rebalancing Agent	Recalculate safety stock, reallocate inventory across DCs, trigger procurement from alternates, within budget/policy guardrails	Takes stockout risk (from Impact Prediction) + current stock (from World State) and proposes a transfer or procurement action, respecting a hard-coded budget/policy guardrail table.	None needed — operates on internal World State inventory data (synthetic for the demo).

Compliance & Audit Agent	Log every autonomous decision with rationale/timestamps/sources; ensure trade-regulation/sanctions/policy compliance; generate audit-ready reports	Houses the **Regulatory & Permission Automation** module (Section 7): sanctions screening via a real free API, cached+scraped import-permit lookups, auto-generated permit requests, and a full timestamped audit log of every decision any agent made.	**Trade.gov Consolidated Screening List API** (free sanctions/denied-party screening, no key: https://developer.trade.gov/consolidated-screening-list.html)

Human-in-the-Loop (mandatory control layer, not a 6th agent): risk-tiered gate — low-risk auto-executes, medium needs one-click confirm, high-risk requires a named human approver with full context shown. No external API — this is a UI + risk-threshold rule.

TIER 1 — Our differentiator modules (live inside the Tier 0 agents above, but built and demoed as distinct, visible capabilities)

Module	Lives inside	What it does	Useful free API

Disruption Verification Layer	Disruption Sensing	Cross-checks any scraped/secondary "war/protest/blockage" claim against ACLED/GDACS/ReliefWeb before it's allowed to influence a route. Confidence-scored: HIGH (structured source) / MEDIUM (2 corroborating sources) / LOW (shown as "unverified," not actioned).	**ACLED** (https://acleddata.com/) as primary corroboration source, plus **ReliefWeb API** (https://apidoc.reliefweb.int/) for humanitarian-access corroboration
Rerouting Intelligence	Rerouting & Logistics	Fuses live traffic (TomTom free tier or OpenRouteService), weather (Open-Meteo), and verified unrest risk into a transparent weighted score per candidate route — explainable on screen, not a black box.	Same as Rerouting & Logistics above: **TomTom** or **OpenRouteService** + **Open-Meteo**
Regulatory & Permission Automation	Compliance & Audit	On every route change: diffs old vs. new regions → cache lookup → scoped scraping on cache-miss → auto-generates and routes the permit request → tracks REQUESTED → PENDING → APPROVED, pausing the shipment at the	**Trade.gov CSL API** for the sanctions-check half (https://developer.trade.gov/consolidated-screening-list.html); no free API exists for country-specific import-permit rules, so this half

		border if a mandatory permit is still pending.	uses the cache-first design in Section 9
Edge/Offline Resilience Layer	Cross-cutting — wraps Disruption Sensing + Rerouting logic	Runs the same excursion-detection and routing rules on a local SQLite store when connectivity drops; syncs and reconciles automatically when it returns. This is your single best live demo moment.	None needed — this runs entirely on local/simulated data by design; it's meant to prove the system works with zero external calls.

TIER 2 — The full supporting roster (restored, all built to at least "working-but-simple" or "minimum-viable-real" depth — nothing here is a stub with no logic)

	Agent	Depth	Minimum real implementation	Useful free API
	Supplier Data Agent	Working-but-simple	One FastAPI route returning synthetic supplier capacity/lead-time JSON, scoped by an API key/role — demonstrates the "data stays with its	None needed — synthetic data behind your own mock endpoint.

			owner" pattern even though the data is synthetic.	
	Logistics Data Agent	Working-but-simple	Same pattern — synthetic shipment/GPS/ETA data behind a scoped endpoint.	None needed for the mock data itself; can optionally use the same **TomTom/OpenRouteService** ETA data as Rerouting & Logistics to make the synthetic ETAs more realistic.
	Warehouse Data Agent	Working-but-simple	Same pattern — synthetic stock-level/temperature data behind a scoped endpoint.	None needed — synthetic data behind your own mock endpoint.
	Cold-Chain Monitoring Agent	Full-depth (this is core to your pharm	A real streaming loop (simulated sensor, but real code: a Python generator pushing timestamped temperature/GPS readings) with a real	**Open-Meteo** can supply real ambient temperature at the shipment's simulated location as a realistic external baseline to layer your simulated cold-chain sensor noise on top of, instead of pure random numbers.

		a angle — don't shortcut it)	rule engine (`if temp > 8°C for >15min → EXCURSION`) writing real state transitions to World State. This is what feeds the edge/offline demo.	
	Demand Forecasting Agent	Working-but-simple	A single scikit-learn regression (or even a seasonal moving-average) on synthetic historical demand by product/region — real model, real predict() call, honestly described as trained on synthetic data.	**WHO Global Health Observatory (GHO) OData API** (free, no key: https://ghoapi.azureedge.net/api, docs at https://www.who.int/data/gho/info/gho-odata-api) — gives you real immunization-coverage and disease-burden indicators per country/year, which is a legitimate real-world proxy to ground your synthetic demand curves rather than inventing them from nothing. Note: WHO is migrating this to a newer OData interface, so double-check the endpoint is live when you build.

	Impact / Stockout Prediction Agent	Working-but-simple	Combines predicted delay + demand forecast + current inventory into a stockout probability via a simple formula or logistic regression — feeds Scenario Planning directly.	None needed — internal computation over other agents' outputs.
	Optimization / Decision Agent	Working-but-simple	Combines Scenario Planning's scored options + Compliance status into a ranked recommendation. A transparent weighted-sum or a real OR-Tools model if someone on the team already knows it — either is legitimate,	None needed — internal computation, and OR-Tools (if used) is a free open-source library, not an API.

			but the weighted-sum is easier to explain live and just as convincing.	
	Monitoring Agent	Working-but-simple	After execution, polls/compares predicted vs. actual (simulated) outcome — predicted ETA vs actual, predicted temp risk vs actual reading — and flags deviation.	None needed — compares your own agents' internal predictions vs. your own simulated "actuals."
	Learning / Calibration Agent	Minimum-viable-real	Logs every (prediction, actual outcome) pair to a table with computed error. Doesn't need to retrain live — say explicitly: "collects prediction-outcome pairs and supports	None needed — internal logging over your own data.

			scheduled recalibration," which is honest and still a real, running piece of code (a logging + error-computation job), not vaporware.	
	Distribution Shift Detector	Minimum-viable-real	A real, simple statistical check: compare the distribution of current input features (e.g., disruption frequency, delay magnitude) against a stored baseline from "normal" synthetic history, using something as simple as a z-score or KS-test. When it fires, it does something	None needed — a statistical test over your own historical vs. live feature data.

			visible: raises the human-approval threshold and flags "operating outside training conditions" on the dashboard. This is a strong, differentiating, and genuinely buildable responsible-AI feature.	
	Digital Twin / What-If Simulation Agent	Minimum-viable-real, scoped to one query pattern	Don't build a general simulation engine — build one real, working query: "what happens if [this warehouse] is unavailable for [N hours]?" that re-runs Impact Prediction + Scenario Planning against a hypothetical World	None needed — it re-runs your own agents against a hypothetical state; no new external data source required.

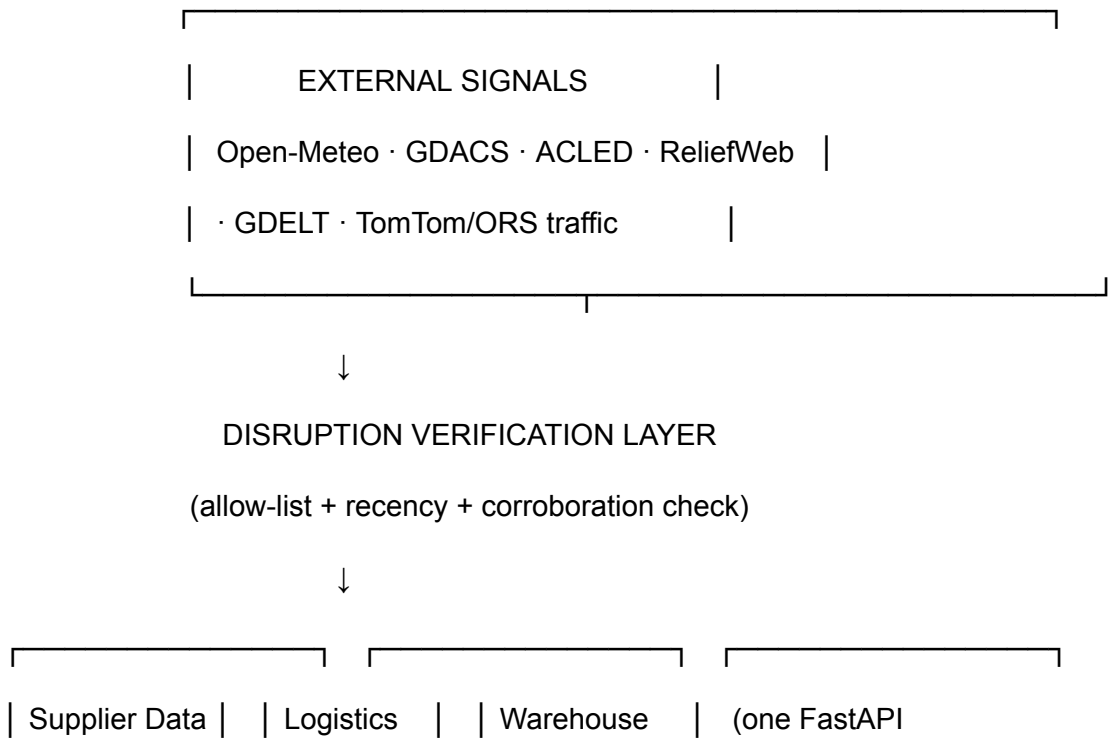
			State snapshot and returns real numbers (affected shipments, expected stockout change, alternate facility suggestion). It's real, it runs, it's just scoped to the one what-if pattern you demo.	
	Cascading Impact Analysis Agent	Minimum-viable-real	Build the supply-chain graph in NetworkX (warehouse → shipment → distributor → clinic edges from your synthetic data) and run one real traversal: given a failed node, return the downstream nodes reachable within N	**HDX (Humanitarian Data Exchange) API** (free, https://data.humdata.org/, docs: https://publicapi.dev/humanitarian-data-exchange-api) — has real, free health-facility location datasets and WorldPop population layers per country, which you can use to tag your synthetic clinics with realistic "population served" numbers so your patient-impact weighting is grounded in real data, not invented.

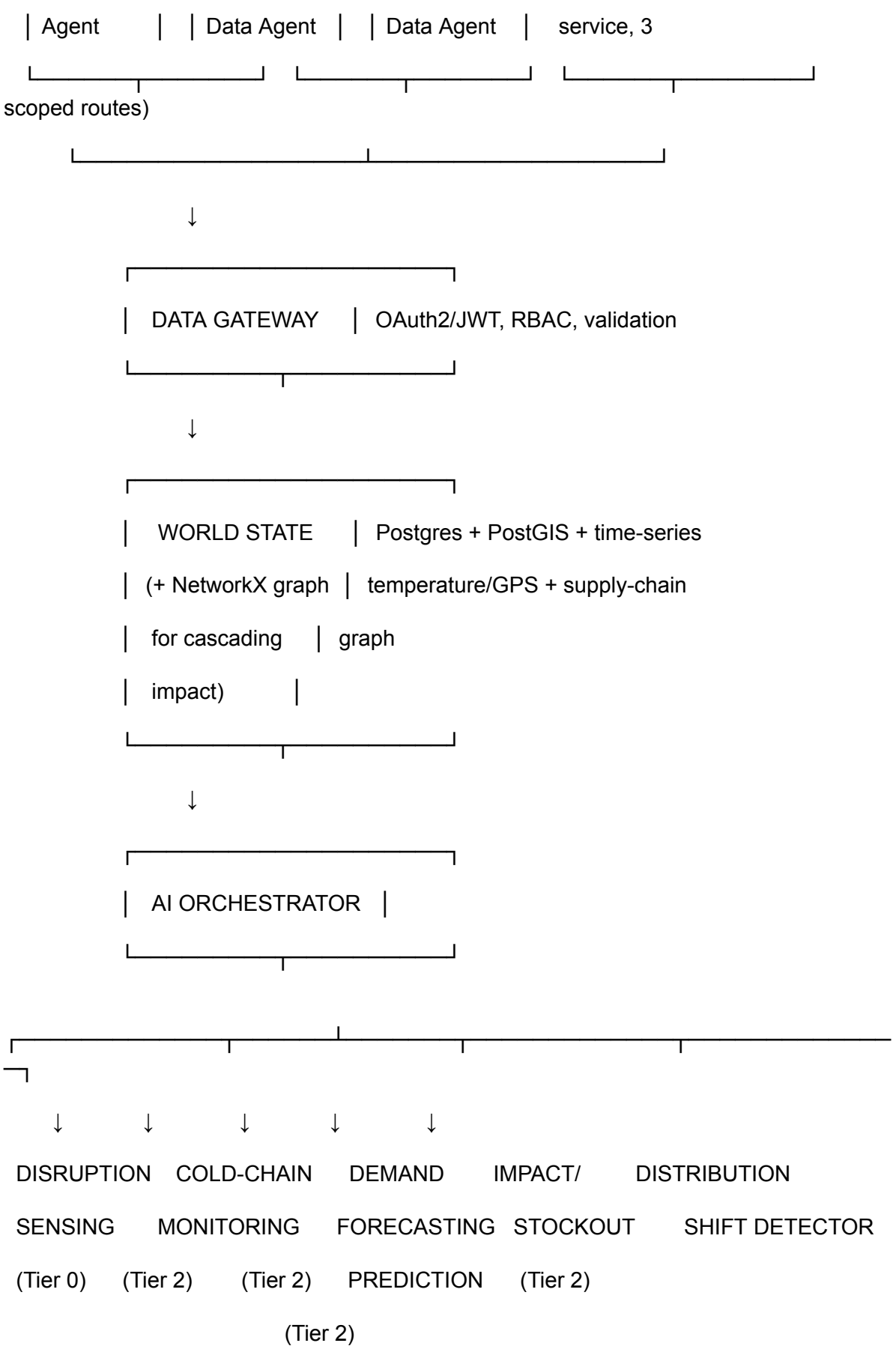
			hops, sorted by patient-impact weight if you tag clinics with population served. This is a real graph algorithm on real (synthetic) data, not a mockup.	

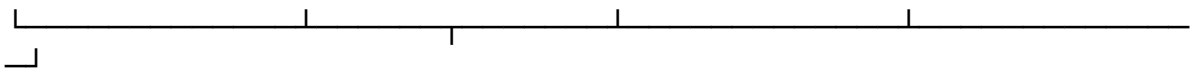
What this gets you: every single agent from your original vision is present, and every single one does something real when you click it — the difference in depth is honest and defensible ("this one's a full production-grade module, this one's a scoped MVP of a bigger idea") rather than invisible ("half of these are just names on a diagram," which is what judges assume by default and will ask you to prove otherwise on the spot).

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4. Full architecture diagram




Agent
Data Agent
Data Agent
service, 3
scoped routes)
DATA GATEWAY
OAuth2/JWT, RBAC, validation
WORLD STATE
Postgres + PostGIS + time-series
(+ NetworkX graph
temperature/GPS + supply-chain
for cascading
graph
impact)
AI ORCHESTRATOR
DISRUPTION SENSING (Tier 0)
COLD-CHAIN MONITORING (Tier 2)
DEMAND FORECASTING (Tier 2)
IMPACT/ STOCKOUT PREDICTION (Tier 2)
DISTRIBUTION SHIFT DETECTOR (Tier 2)



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SCENARIO PLANNING (Tier 0)

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DIGITAL TWIN / WHAT-IF (on demand, Tier 2)

CASCADING IMPACT ANALYSIS (on demand, Tier 2)

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COMPLIANCE & AUDIT (Tier 0)

[Regulatory & Permission Automation]

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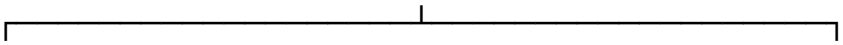
OPTIMIZATION / DECISION (Tier 2)

↓

HUMAN APPROVAL GATEWAY (mandatory)

risk-tiered: auto / confirm / mandatory

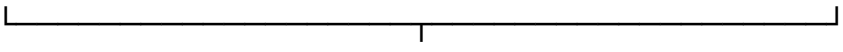
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REROUTING & LOGISTICS (Tier 0) INVENTORY REBALANCING (Tier 0)

[Rerouting Intelligence]



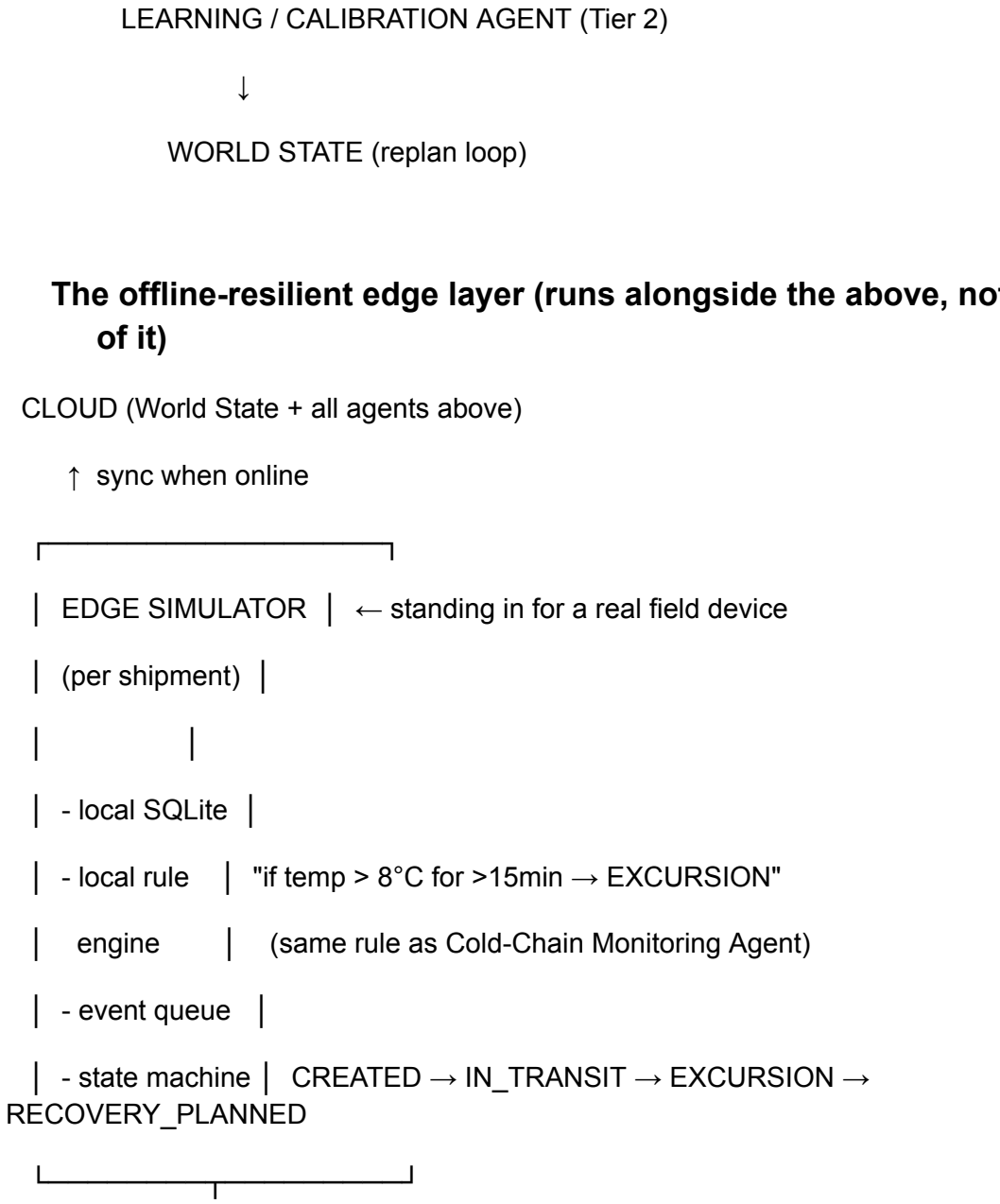
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EXECUTE

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MONITORING AGENT (Tier 2)

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The offline-resilient edge layer (runs alongside the above, not instead of it)

CLOUD (World State + all agents above)

↑ sync when online

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│ EDGE SIMULATOR │ ← standing in for a real field device

│ (per shipment) │

│ │

│ - local SQLite │

│ - local rule │ "if temp > 8°C for >15min → EXCURSION"

│ engine │ (same rule as Cold-Chain Monitoring Agent)

│ - event queue │

│ - state machine │ CREATED → IN_TRANSIT → EXCURSION → RECOVERY_PLANNED

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│ connectivity toggled OFF in the demo

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keeps logging + deciding locally, zero cloud calls

│ connectivity toggled ON

↓

SYNC ENGINE → bulk-uploads queued events → World State recomputes →

dashboard shows "N events synchronized" and updated risk

Buildable in pure software: simulate the temperature/GPS stream, simulate a "connectivity" boolean you flip live in the demo, and use SQLite + timestamp-based/last-write-wins reconciliation on sync (a standard, documented offline-first pattern — no need for CRDTs at hackathon scale).

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5. Rerouting Intelligence — weather + traffic + verified unrest, in detail

Inputs fused:

Signal	Source	What it tells the agent
Traffic	TomTom free tier (real live traffic) or OpenRouteService (free, graph-based ETA, no live traffic)	Congestion, ETA per candidate route
Weather	Open-Meteo (free, no key)	Storms, flooding, extreme heat, visibility along waypoints
Civil unrest / blockage	Disruption Verification Layer output only — never raw scraped text	Whether a road/border/port segment is currently unsafe or blocked

Scoring (transparent, explainable on screen):

- route_score = w1 * traffic_delay_minutes
- + w2 * weather_risk_score (0–1)
- + w3 * unrest_risk_score (0–1, verified signals only)
- + w4 * temperature_exposure_risk (longer/hotter route = more cold-chain risk)

A plain weighted sum beats a black-box model here — you can show judges *why* Route B beat Route A, live.

Flow: candidate routes generated → each scored → best-scoring route (if different from current) is proposed to Scenario Planning → still goes through Compliance + Optimization + Human Approval. This agent proposes; it never silently reroutes.

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6. Disruption Verification Layer, in detail

Raw scraping of arbitrary news for "war" or "protest" produces false positives (old articles, satire, rumor) that could wrongly reroute a life-saving shipment. So this is a verification pipeline, not a scraper:

Trusted, structured sources first:

- **ACLED** — dated, geolocated, sourced records of political violence/protest in 200+ countries. Primary unrest signal.
- **GDACS** — natural-disaster alerts.
- **ReliefWeb/OCHA** — humanitarian-crisis and access-constraint updates.
- **Government travel advisories** — already curated by a government; safe to treat as trusted.

Scraping only supplementary, from a small pre-defined allow-list of official domains — never an open crawl.

Verification logic:

New signal arrives (scraped or secondary feed)

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1. Source allow-list check → reject if not on the trusted domain list

2. Recency check → reject if stale (e.g. >48h for fast-moving events)

3. Corroboration check → require agreement with a structured source (ACLED/GDACS/ReliefWeb) OR two independent allow-listed sources

4. Confidence assigned → HIGH (structured) / MEDIUM (2 corroborating) / LOW (single, unactioned)

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Only HIGH/MEDIUM reach Rerouting Intelligence.

LOW shows in the dashboard as "unverified — under review," never actioned.

Judge-facing line: *"We don't reroute a vaccine shipment because of one unverified rumor — we require corroboration from a structured, trusted dataset."*

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7. Regulatory & Permission Automation, in detail

Triggered whenever a route changes:

Approved route change (old → new)

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Compute NEW regions the route now passes through

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For each new region:

1. CACHE LOOKUP — Postgres table: region → permits/templates/authority/last-verified date

hit → use cached rules, flagged with "last verified: <date>"

miss/stale → 2. SCOPED SCRAPE of that region's official customs/import portal (allow-listed only)

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Update cache with fresh result + timestamp

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For each required permit:

check if shipment already has a valid one on file

no/expired → auto-generate request (pre-filled from World State: shipment ID,

product, quantity, origin, destination, batch)

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route to the region's issuing authority (if e-submission exists) or the internal

compliance officer for manual filing

↓

track REQUESTED → PENDING → APPROVED/REJECTED

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if a MANDATORY permit is still PENDING when the shipment reaches that border →

pause the shipment and notify Human Approval Gateway (never proceed silently)

Sanctions/denied-party screening within this same agent uses the **Trade.gov Consolidated Screening List API** (Section 9) — free, structured, no scraping needed for that specific check.

Dashboard Timeline — every event above writes to a single chronological feed per shipment:

- 14:02 Disruption detected — port congestion, verified (HIGH confidence, ACLED + GDACS)
- 14:03 Rerouting & Logistics: Route B recommended (traffic ETA -1.5h, unrest risk 0.0 vs 0.6)
- 14:04 Compliance & Audit: Route B enters Country C (new jurisdiction)
- 14:04 Cache hit — Country C requires Import Permit + Cold-Chain Declaration
- 14:05 Permission request auto-generated, sent to Country C Customs Authority
- 14:05 Timeline: "Import Permit — PENDING"
- 15:40 Timeline: "Import Permit — APPROVED"
- 15:41 Shipment cleared to proceed on Route B

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8. Why this is genuinely current, not just plausible-sounding

- **Agentic LLMs for supply-chain coordination and disruption monitoring are an active 2025–2026 research area.** Brintrup, Jannelli, Schoepf and colleagues have published on <cite index="9-1">using LLM agents to automate coordination decisions like inventory levels and delivery times across supply chain tiers</cite>, and a follow-up specifically on <cite index="8-1">a minimally supervised, seven-agent architecture for monitoring and responding to disruptions across multi-tier supply networks, aimed at shifting from reactive recovery to proactive resilience</cite>. Cite this directly — it validates your multi-agent framing.
- **The same research stream flags exactly the failure mode your design guards against**: agentic systems can <cite index="5-1">inherit managerial biases and produce confidently wrong decisions without explicit verification</cite>. That's precisely why compliance/sanctions/safety checks here are rule-based, not model-based, and why Human-in-the-Loop is mandatory on high-stakes actions.
- **Offline-first design for underserved, low-connectivity regions is a mature, active engineering pattern**, not speculative. Field-health practitioners note there are

roughly <cite index="16-1">3.5 billion people with little to no reliable internet access, concentrated in rural India, Sub-Saharan Africa, Southeast Asia, and South America — the same places with the highest preventable-disease burden</cite>, and standard practice treats disconnection as normal, <cite index="14-1">resolving conflicts automatically or with lightweight rules once devices reconnect</cite>, with approaches as simple as <cite index="20-1">timestamp-based or last-write-wins conflict resolution</cite> being sufficient for most field-data use cases.

Two-sentence "why now" for judges: *agentic multi-tier supply chain coordination is an active research area right now, and offline-first field architecture is a mature pattern for underserved regions — nobody has combined them for pharma cold-chain specifically. That gap is your project.*

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9. API & data-source appendix — free options only, with links

Traffic / routing

API	Free?	Link	Notes
OpenRouteService	Yes, free key, no card	https://openrouteservice.org/dev/#/api-docs	No live traffic layer — route geometry/ETA only
TomTom Routing + Traffic	Yes, free daily quota, no card for evaluation	https://developer.tomtom.com/traffic-api/documentation	Genuine live traffic on the free tier — use this for the "TRAFFIC_AWARE" demo moment
Google Maps Routes API	Requires card-linked billing (free credits only)	https://developers.google.com/maps/documentation/routes	Best quality, but not truly free — optional/stretch only

OSRM	Yes, fully free, public demo server	http://project-osrm.org/	Fallback if ORS rate-limits mid-demo

Weather

API	Free?	Link
Open-Meteo	Yes, completely free, no key, no signup	https://open-meteo.com/

Conflict / civil unrest / disaster

API	Free?	Link
ACLED	Yes, free with registration	https://acleddata.com/data-export-tool/
GDACS	Yes, free, public	https://www.gdacs.org/xml/rss.xml
ReliefWeb (OCHA)	Yes, free, no key for basic queries	https://apidoc.reliefweb.int/
GDELT	Yes, completely free, no key	https://www.gdeltproject.org/data.html

Sanctions / denied-party screening — replaces scraping entirely

API	Free?	Link

Trade.gov Consolidated Screening List	Yes, free, no key required	https://developer.trade.gov/consolidated-screening-list.html

Public health context data (for Demand Forecasting & Cascading Impact Analysis)

API	Free?	Link	Used for
WHO Global Health Observatory (GHO) OData API	Yes, free, no key	https://ghoapi.azureedge.net/api (docs: https://www.who.int/data/gho/info/gho-odata-api)	Real immunization-coverage/disease-burden indicators to ground synthetic demand curves. Note: WHO is migrating this API — verify the endpoint is live when you build.
HDX (Humanitarian Data Exchange)	Yes, free (registration for some datasets)	https://data.humdata.org/ (API docs: https://publicapi.dev/humanitarian-data-exchange-api)	Real health-facility location and WorldPop population datasets — use to tag your synthetic clinics with realistic "population

			served" for the Cascading Impact Analysis Agent's patient-weighting.

Country-specific import/customs permit rules — the one honest gap

No single free, structured, global API tells you "pharma import permit requirements for Country X today." WTO/UN Comtrade cover tariffs/trade flows, not permit documents. Build the **cache-first Postgres table** seeded by hand for your 4–6 demo countries, with scoped, allow-listed scraping of official customs portals as the documented fallback for cache-misses. Say this openly to judges — it's the honest and defensible way to handle a gap that genuinely exists in the data ecosystem.

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10. Demo script — showcasing the full roster in ~6 minutes

1. **Setup (20s):** Dashboard shows Medicine A moving from a regional cold-storage hub toward a rural clinic; temperature/location streaming from the Cold-Chain Monitoring Agent.
2. **Disruption + verification (30s):** Inject a port disruption. Disruption Verification Layer shows it HIGH confidence (ACLED+GDACS corroborated) vs. a second unverified claim shown as "under review, not actioned."
3. **Cascading impact (20s):** Trigger Cascading Impact Analysis — show the graph traversal: this warehouse failure reaches 3 downstream clinics within 2 hops, prioritized by patients served.
4. **Digital twin what-if (20s):** Ask the dashboard "what if this warehouse is down for 6 hours?" — Digital Twin Agent reruns Impact Prediction against the hypothetical state and returns real numbers live.
5. **Rerouting intelligence (30s):** Rerouting & Logistics scores 2–3 candidate routes on live traffic + weather + verified unrest, picks the winner with a visible breakdown.
6. **Regulatory auto-response (30s):** Compliance & Audit detects the new jurisdiction, cache-hits the permit requirements, auto-generates the request, posts it to the Dashboard Timeline as PENDING.
7. **Scenario + optimization (20s):** Scenario Planning + Optimization present the ranked plan with cost/risk; Compliance flags one option as non-viable.
8. **Human approval (15s):** Risk-tiered gate — high-risk action requires a named approver.

9. **Distribution shift (15s):** Briefly show the Distribution Shift Detector flagging that current conditions are outside training data, raising the approval threshold automatically.
10. **The differentiator — offline resilience (60s, spend the most time here):** Kill connectivity live. Edge layer keeps detecting the temperature excursion and transitioning state with zero cloud calls. Restore connectivity — "47 events synchronized," dashboard catches up instantly.
11. **Close (20s):** "Every one of these agents is real and running — not a slide. And it works the same way whether the shipment is in Chennai or in a village with no signal."

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11. Round 1 framing (Design Thinking canvas)

- **Stakeholders to empathy-map:** rural clinic health worker, logistics planner, compliance officer, smallholder/last-mile distributor, patient.
- **Failure modes to name explicitly:** cold-chain break → medicine destroyed (not delayed); connectivity loss → invisible shipment → silent failure; regulatory mismatch on reroute → shipment stuck at a border; single-node failure → cascading multi-clinic shortage.
- **Business model angle:** position as SAP Business Network/IBP-adjacent tooling — a resilience layer plugging into existing ERP/TMS via the data-agent pattern, sellable to pharma distributors and NGO/public-health logistics operators (e.g., Gavi-style vaccine networks), with an accessibility-priced tier for public-health/NGO users — this is what makes it useful to "everyone, not just those who can easily afford it."

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12. What to say precisely (don't overclaim, judges will probe this)

- The system **recommends** route/supplier changes; humans approve anything high-stakes — never claim full autonomy on jurisdiction changes.
- ML components **estimate probability from limited/synthetic data** and improve with real deployment data — don't claim production-grade accuracy.
- SMS/cellular fallback is a **designed communication tier**, not demoed, unless you actually wire up something like Twilio's sandbox.
- The system **auto-generates and auto-routes** permit requests — it does not auto-submit legally binding approvals to governments (most portals have no public API for that).
- Unrest/conflict detection is **corroborated against trusted datasets**, not real-time-perfect — ACLED and similar sources have reporting lag and regional coverage gaps.

- The Digital Twin and Cascading Impact agents are **real but scoped** — one working what-if pattern and one working graph traversal, not a general-purpose simulation engine. Say this proactively; it reads as rigor, not weakness.
- The country-regulation lookup is **cache-first with a narrow scraping fallback for a handful of demo countries** — not full global live coverage, because no clean data source for that exists yet.